



Operations Research

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Exercise Sheet 2

Training Exercises

Exercise T2.1. *Simplex algorithm*

Consider the following linear program (LP):

$$\begin{array}{ll} \text{Max} & x_1 + x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & 2x_1 + x_2 \leq 11 \\ & x_2 \leq 5 \\ & x \in \mathbb{R}_{\geq 0}^2 \end{array}$$

- Solve the LP graphically.
- Convert the LP into standard form.
- Is the LP in canonical form? Justify.
- Solve the LP with the simplex algorithm in tableau notation using the Big M method.
- Is the LP degenerate? Justify your answer. How can you read such degeneracy from the optimal tableau of the simplex algorithm?
- The final tableau of task part (d) is an optimal simplex tableau. Does another optimal simplex tableau exist? Give one or justify why no other optimal simplex tableau can exist.

Exercise T2.2. *Simplex algorithm*

- The following linear program is given in tableau notation. Specify all pivot elements by means of which the objective function value can be increased in the next iteration of the simplex algorithm. For each of these pivot elements, also name the entering and exiting basis variable. Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	-1	-2	1	0	0	0	0
x_4	-1	2	1	1	0	0	3
x_5	2	4	4	0	1	0	4
x_6	-1	0	2	0	0	1	6

- b) The following linear program is given in tableau notation. Here x_1 and x_2 are structure variables, whereas x_3 , x_4 and x_5 are slack variables. Decide whether the underlying linear program is optimally solvable, intractable, or unbounded. If optimal solutions exist, explicitly state all optimal solutions $x^* = (x_1, x_2)$ for the underlying linear program. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	0	3	1	0	1
x_2	0	1	1	1	0	1
x_1	1	0	-1	1	0	1
x_5	0	0	-1	-1	1	0

- c) The following linear program is given in tableau notation. Here x_1 and x_2 are structure variables, whereas x_3 , x_4 and x_5 are slack variables. Decide whether the underlying linear program is optimally solvable, intractable, or unbounded. If optimal solutions exist, explicitly state all optimal solutions $x^* = (x_1, x_2)$ for the underlying linear program. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	0	0	1	0	5
x_3	0	0	1	-2	3	3
x_2	0	1	0	1	-1	2
x_1	1	0	0	0	1	3

- d) The following linear program is given in tableau notation. Here x_1 and x_2 are structure variables, whereas x_3 and x_4 are slack variables. The variable w_1 is an artificial variable, which was introduced to be able to solve the linear program using the Big M method. Decide whether the underlying linear program is optimally solvable, intractable, or unbounded. If optimal solutions exist, explicitly state all optimal solutions $x^* = (x_1, x_2)$ for the underlying linear program. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	w_1	
z	M	0	$1 + M$	M	0	$1 - M$
x_2	1	1	1	0	0	1
w_1	-1	0	-1	-1	1	1

Self-study Exercises

Exercise S2.1. Simplex algorithm

- a) The following linear program is given in tableau notation. In what value ranges must the parameters α and β lie so that in the next iteration of the simplex algorithm the objective function value can be increased by x_1 entering the basis and x_2 leaving the basis? Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	α	0	2	0	3	0	0
x_4	-1	0	-2	1	5	0	1
x_6	2	0	0	0	1	1	4
x_2	6	1	7	0	3	0	β

- b) The following linear program is given in tableau notation. Give the largest possible value ranges for the parameters α and β so that one can read directly from the simplex tableau that the underlying linear program is unbounded. Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	0	3	0	α	1	0	0
x_3	0	-1	1	-1	3	0	1
x_6	0	-4	0	-3	1	1	3
x_1	1	2	0	β	1	0	2

- c) The following linear program is given in tableau notation. Give a value for each of the parameters α and β such that degenerate basis solutions exist. Justify your answer.

Base	Coefficients						Res.
	x_1	x_2	x_3	x_4	x_5	x_6	
z	-1	-4	-1	0	0	0	0
x_4	-1	2	β	1	0	0	4
x_5	2	1	2	0	1	0	9
x_6	α	6	3	0	0	1	12

Exercise S2.2. Big M method

Consider the following linear program (LP):

$$\begin{aligned}
 \text{Max} \quad & x_1 + x_2 \\
 \text{s.t.} \quad & x_1 + 2x_2 = 4 \\
 & x_1 - x_2 = 1 \\
 & 2x_1 + x_2 \geq 5 \\
 & x \in \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

Compute an optimal basis solution for the LP with the simplex algorithm using the Big M method.

Exercise S2.3. *Simplex tableau special cases*

Consider the following linear programs in tableau representation:

a)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z				0	1	10
x_3			1	2	-1	5
x_2		1		1	0	4
x_1	1			-2	1	2

b)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-50	-45				0
x_3	6	5	1			60
x_4	6	5		1		0
x_5	10	20			1	150

c)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	2	-1	0			2
x_3	-2	-1	1			2
x_4	1	0		1		4
x_5	0	-2			1	1

d)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-50	-45				0
x_3	6	5	1			60
x_4	16	25		1		320
x_5	10	20			1	150

e)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-7	-5				0
x_3	3	4	1			24
x_4	2	1		1		10
x_5	-1				1	-1

f)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-7	-5				0
x_3	3	4	1			24
x_4	2	1		1		10
x_5	1				1	-1

Indicate in each case which special case it is and how to recognize this special case in the tableau representation.

Furthermore, indicate how the respective special case affects the flow of the simplex algorithm.

Exercise S2.4. *Simplex tableau general and special cases*

a) Transform the following LP into canonical form.

$$\begin{aligned}
 \text{Max} \quad & x_1 + 2x_2 \\
 \text{s.t.} \quad & -2x_1 + x_2 \geq -2 \\
 & -x_1 + x_2 \leq 3 \\
 & x_1 + x_2 \leq 7 \\
 & x \in \mathbb{R}_{\geq 0}^2
 \end{aligned}$$

b) Perform one iteration of the simplex algorithm. State the incoming and outgoing basic variables.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-3			2		6
x_3	1		1	1		5
x_2	-1	1		1		3
x_5	2			-1	1	4

Is the resulting tableau optimal?

c) For the following LP in tableau notation, provide the largest possible ranges for α and β such that one can read directly from the simplex tableau that the underlying LP is unbounded. Justify your answer.

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	α			2		6
x_3	-1		1	1		5
x_2	-1	1		1		3
x_5	β			1	-1	-4

d) Can the same vertex be visited more than once during the simplex algorithm? Justify your answer.

e) Can a vertex be revisited after a different vertex was visited? Justify your answer.